

A noncanonical automorphism at the critical polynomial weight

Automated mathematics run `fa_banach_001`

Model: GPT5.6

9 August 2026

Status. Claimed full negative resolution (counterexample), likely valid pending human review. The construction works for every $1 < p < \infty$.

1 Source question

Kuznetsova and Zadeh study algebra automorphisms of $\ell^p(\mathbb{Z}, \omega_a)$, where

$$\omega_a(n) = \max(1, |n|^a).$$

After proving the canonical classification in their higher-regularity range, they state on published page 864, in Section 3 immediately after the relevant corollary: “For $a = 1$ and $p > 1$, the question is open” [1, p. 864]. In context, the question is whether every automorphism at the critical weight ω_1 must still be canonical, namely of one of the forms

$$(Tf)_n = \lambda^n f_n, \quad (Tf)_n = \lambda^n f_{-n} \quad (\lambda \in \mathbb{T}).$$

864

Y. Kuznetsova and S. Zadeh

If $p > 1$ and $a \geq 2$, the result follows from a generalisation of the Beurling-Helson theorem given by Lebedev and Oleviskii [LO]: if $\varphi : \mathbb{T} \rightarrow \mathbb{T}$ is a C^1 function of the argument and the composition operator given by φ is bounded on $\ell^p(\mathbb{Z})$, then φ is a monomial. In our case, $\varphi \in \ell^p(\mathbb{Z}, \omega_a)$, which implies $\varphi \in C^1$. The rest is proved as in the ℓ^1 case. ■

For $a = 1$ and $p > 1$, the question is open. For fast growing weights, we can observe a similar behaviour.

Example 3.3 Let w be a weight on $\mathbb{Z}$ of one of the two forms: $w_n = e^{n^\gamma}$, $0 < \gamma < 1$, or $w_n = a^{|n|}(1 + n^2)$, $a > 1$. Then the elementary Blaschke product $b_r(z) = \frac{z-r}{1-\bar{r}z}$, $r > 0$,

Figure 1: The open statement and its immediate context in [1, p. 864].

2 Definitions

Write $\mathbb{T} = \mathbb{R}/\mathbb{Z}$ and $e_n(x) = e^{2\pi i n x}$. For $1 < p < \infty$, let

$$\mathcal{FL}^p(\mathbb{T}) = \left\{ u : (\widehat{u}(n))_{n \in \mathbb{Z}} \in \ell^p(\mathbb{Z}) \right\}, \quad \|u\|_{\mathcal{FL}^p} = \|\widehat{u}\|_{\ell^p},$$

initially in the sense of periodic distributions. The Fourier transform identifies $\ell^p(\mathbb{Z}, \omega_1)$ with the completion A_p^1 of the trigonometric polynomials in the norm

$$\|f\|_{A_p^1}^p = |\widehat{f}(0)|^p + \sum_{n \neq 0} |n \widehat{f}(n)|^p. \quad (1)$$

Moreover, convolution of sequences corresponds to pointwise multiplication in A_p^1 .

If $q = p/(p-1)$, Hölder's inequality gives

$$\|f\|_\infty \leq \sum_n |\widehat{f}(n)| \leq |\widehat{f}(0)| + \left(2 \sum_{n \geq 1} n^{-q} \right)^{1/q} \left(\sum_{n \neq 0} |n \widehat{f}(n)|^p \right)^{1/p}. \quad (2)$$

Thus A_p^1 embeds continuously into $C(\mathbb{T})$.

3 Idea of the proof

At order two, the source paper obtains C^1 regularity and invokes a Beurling–Helson rigidity theorem. At the critical first-order norm, corners are still allowed. We exploit precisely that endpoint: a rational piecewise-affine homeomorphism has constant derivative on each branch, while the branch cutoffs are bounded-variation multipliers on $\mathcal{FL}^p$.

For a rational slope r/s , splitting Fourier frequencies modulo s turns a localized affine composition into finitely many sparse integer-frequency reindexings. Each is followed only by multiplication by a bounded-variation function. Such multiplication is bounded on $\mathcal{FL}^p$ because, after one integration by parts, its coefficient convolution is an average of modulated discrete Hilbert transforms. Applying this to the derivative of a composed trigonometric polynomial controls the critical weighted norm. The same argument applies to the rational piecewise-affine inverse.

4 Two Fourier lemmas

Lemma 1 (bounded-variation multipliers). *Let $1 < p < \infty$ and let m be a complex-valued function of bounded variation on $\mathbb{T}$. Then multiplication by m is bounded on $\mathcal{FL}^p(\mathbb{T})$.*

Proof. Let $\mu = Dm$ be the finite complex measure which is the distributional derivative of m . With the normalization above, for $k \neq 0$,

$$2\pi i k \widehat{m}(k) = \widehat{\mu}(k).$$

For a finitely supported sequence a , convolution by the nonzero Fourier coefficients of m can therefore be written as

$$\sum_{k \neq 0} \widehat{m}(k) a_{n-k} = \frac{1}{2\pi i} \int_{\mathbb{T}} \left(\sum_{k \neq 0} \frac{e^{-2\pi i k t}}{k} a_{n-k} \right) d\mu(t).$$

For fixed t , the operator in parentheses is a modulation conjugate of the discrete Hilbert transform

$$Ha(n) = \sum_{k \neq 0} \frac{a_{n-k}}{k}.$$

The discrete Hilbert transform is bounded on $\ell^p(\mathbb{Z})$ for $1 < p < \infty$; this is the standard discrete singular-integral theorem (see, for example, the transference treatment in [3]). All modulation conjugates have the same norm. Minkowski's integral inequality now bounds the displayed expression by $C_p \|\mu\| \|a\|_{\ell^p}$. The $k = 0$ term is $\widehat{m}(0)a_n$. Hence convolution by $\widehat{m}$, equivalently multiplication by m on $\mathcal{FL}^p$, is bounded. $\square$

Lemma 2 (a localized rational affine branch). *Let $I \subset \mathbb{T}$ be an interval, let r, s be positive integers, and let $b \in \mathbb{R}$. On trigonometric polynomials define*

$$S_{I,r/s,b}f(x) = \mathbf{1}_I(x)f\left(\frac{r}{s}x + b\right).$$

Then $S_{I,r/s,b}$ extends boundedly on $\mathcal{FL}^p(\mathbb{T})$ for every $1 < p < \infty$.

Proof. It is enough to take r/s in lowest terms. If $f(x) = \sum_n a_n e_n(x)$, split $n = sk + j$ with $0 \leq j < s$. Directly,

$$\mathbf{1}_I(x)f\left(\frac{r}{s}x + b\right) = \sum_{j=0}^{s-1} m_j(x)F_j(x), \quad (3)$$

where

$$m_j(x) = \mathbf{1}_I(x)e^{2\pi i r j x / s}, \quad F_j(x) = \sum_{k \in \mathbb{Z}} a_{sk+j} e^{2\pi i (sk+j)b} e_{rk}(x).$$

The Fourier coefficients of F_j are just the j th residue class of (a_n) , inserted at the multiples of r and multiplied by unimodular numbers. Consequently

$$\|F_j\|_{\mathcal{FL}^p}^p = \sum_k |a_{sk+j}|^p.$$

Each m_j , viewed periodically, has bounded variation: it is smooth on I and zero off I , with only finitely many endpoint jumps. Lemma 1, followed by the finite sum in (3), proves the claim. $\square$

5 The counterexample

Theorem 1. *For every $1 < p < \infty$, the convolution Banach algebra $\ell^p(\mathbb{Z}, \omega_1)$ has a noncanonical algebra automorphism. More precisely, extend*

$$\Phi(x) = \begin{cases} 2x, & 0 \leq x \leq \frac{1}{3}, \\ \frac{x+1}{2}, & \frac{1}{3} \leq x \leq 1 \end{cases} \quad \text{by} \quad \Phi(x+1) = \Phi(x) + 1.$$

Then $\phi(x + \mathbb{Z}) = \Phi(x) + \mathbb{Z}$ is a non-affine circle homeomorphism, and composition $C_\phi f = f \circ \phi$ is a bounded invertible algebra automorphism of A_p^1 . Conjugating by the Fourier transform gives the claimed automorphism of $\ell^p(\mathbb{Z}, \omega_1)$.

Proof. The two formulas for Φ agree at $x = 1/3$, and $\Phi(0) = 0$, $\Phi(1) = 1$. Thus the periodic lift is strictly increasing and induces an orientation-preserving circle homeomorphism. Its inverse lift is

$$\Psi(y) = \begin{cases} y/2, & 0 \leq y \leq \frac{2}{3}, \\ 2y - 1, & \frac{2}{3} \leq y \leq 1, \end{cases} \quad \Psi(y+1) = \Psi(y) + 1.$$

Let f be a trigonometric polynomial and set $g = f \circ \phi$. The function g is continuous and piecewise smooth. Since it is continuous at every breakpoint, its distributional derivative has no point masses. The usual chain rule on the two open branches therefore gives

$$g' = 2\mathbf{1}_{[0,1/3]}(f' \circ (2x)) + \frac{1}{2}\mathbf{1}_{[1/3,1]}(f' \circ ((x+1)/2)). \quad (4)$$

Both summands are instances of Lemma 2, with slopes 2 and 1/2. Hence

$$\|g'\|_{\mathcal{FL}^p} \leq C_p \|f'\|_{\mathcal{FL}^p}. \quad (5)$$

The mean coefficient is controlled using (2):

$$|\widehat{g}(0)| \leq \|g\|_\infty = \|f\|_\infty \leq C'_p \|f\|_{A_p^1}.$$

Since $\widehat{f}'(n) = 2\pi i n \widehat{f}(n)$, equations (1) and (5) show that

$$\|C_\phi f\|_{A_p^1} \leq C''_p \|f\|_{A_p^1}.$$

Trigonometric polynomials are dense in A_p^1 , so C_ϕ extends to a bounded operator on A_p^1 . The embedding (2) shows that convergence in A_p^1 implies uniform convergence. Thus this extension is still literal pointwise composition by ϕ , rather than merely an abstract operator defined by completion.

The inverse lift Ψ is also rational piecewise affine. Repeating the same argument proves that $C_{\phi^{-1}}$ is bounded on A_p^1 . On polynomials, and then by density,

$$C_{\phi^{-1}}C_\phi = C_\phi C_{\phi^{-1}} = I.$$

Composition preserves products, so C_ϕ is a bounded algebra automorphism. Under Fourier transform, multiplication in A_p^1 is convolution in $\ell^p(\mathbb{Z}, \omega_1)$, proving the algebra assertion.

Finally, a canonical automorphism has circle symbol $x \mapsto x + c$ or $x \mapsto -x + c$. Our symbol is orientation preserving but has slopes 2 and 1/2 on sets of positive measure, so it is neither. The automorphism is noncanonical. $\square$

Remark 1. *The construction actually proves a small general statement: every circle homeomorphism whose lift and inverse are piecewise affine with finitely many rational slopes induces an automorphism of A_p^1 for $1 < p < \infty$. Only the explicit two-branch example is needed for the source question.*

6 Verification and limitations

- The argument covers every $1 < p < \infty$. It does not cover $p = 1$, because the discrete Hilbert transform is not bounded on ℓ^1 ; this is consistent with classical Beurling–Helson rigidity.
- The derivative formula is first used only for trigonometric polynomials. Continuity at the breakpoints removes delta masses, and the bounded operator is then obtained by completion. This avoids assuming that an arbitrary $\mathcal{FL}^p$ derivative is a function.

- A deterministic finite-Fourier smoke test is included in `code/numerical_smoke_test.py`. It checks the explicit formulas and shows stable sample norm ratios, but it is not part of the proof.
- Bounded novelty search on 9 August 2026: the run’s registry, solution, attempt, and proof-gap indexes were searched by arXiv id, title, and core terms. Exact-phrase searches for the open sentence, DOI/title searches, author-plus- $a = 1$ searches, and searches combining piecewise-linear changes with weighted Fourier–Lebesgue spaces found the source and general change-of-variable literature, but no later resolution of this exact question. This is evidence, not a guarantee, of novelty.
- Recommended human check: audit Lemmas 1 and 2, especially the periodic bounded-variation interpretation of $\mathbf{1}_I e^{2\pi i \alpha x}$, and then the completion argument in Theorem 1.

References

- [1] Y. Kuznetsova and S. Zadeh, *On isomorphisms between weighted L^p -algebras*, Canadian Mathematical Bulletin **64** (2021), no. 4, 853–866; arXiv:1803.06083; doi:10.4153/S0008439520000880.
- [2] V. Lebedev and A. Olevskii, *C^1 changes of variable: Beurling–Helson type theorem and Hörmander conjecture on Fourier multipliers*, Geometric and Functional Analysis **4** (1994), no. 2, 213–235.
- [3] L. Grafakos, *Classical Fourier Analysis*, 3rd ed., Graduate Texts in Mathematics 249, Springer, 2014. See the treatment of singular integrals and transference underlying the boundedness of the discrete Hilbert transform on $\ell^p(\mathbb{Z})$, $1 < p < \infty$.